A Self-Contained Interactive Notebook for Information-First Research using the Universal Binary Principle (UBP) v3.6

Euan R. A. Craig
New Zealand
info@digitaleuan.com

November 2025

Abstract

This paper introduces a self-contained, interactive Jupyter notebook for conducting scientific research within the Universal Binary Principle (UBP) framework, version 3.6.

The UBP is a deterministic, information-first model of reality that posits a 12D+ bitfield substrate governed by a computational grammar. A key barrier to the adoption of such novel frameworks is accessibility. To address this, we developed a comprehensive, zero-dependency Python implementation of the UBP 3.6 system, packaged into a single, executable notebook. This notebook requires no external libraries and runs in any standard Jupyter environment, including Google Colab, making the framework widely accessible. We document the notebook's architecture and the integration of critical fixes that render it a stable, patch-free research tool.

To demonstrate its utility, we present a case study: a *Hydro-Informational Twinning Analysis* of water (H_2O) and hydronium (H_3O^+). The study analyzes the informational cost and topological complexity of molecular state changes, revealing that geometric reconfiguration ($H_2O \rightarrow H_3O^+$) is informationally more expensive than phase disorder (liquid $\rightarrow$ vapor). This result provides empirical support for UBP's core tenet that topology is a primary determinant of informational cost. The notebook is made publicly available to encourage further research and validation of the UBP framework.

1 Introduction

The exploration of fundamental physics and computation has led to the development of numerous theoretical frameworks that challenge the standard model. One such framework is my Universal Binary Principle (UBP), a deterministic model that describes reality as fundamentally informational [1]. The UBP proposes that the universe operates on a discrete, toggle-based substrate governed by a computational grammar, where physical laws emerge from geometric and informational constraints - or at least that it can be modeled as such to and extremely high degree of precision.

While novel frameworks like the UBP offer promising new perspectives, their adoption and validation by the broader scientific community are often hindered by a significant accessibility barrier. Complex theoretical underpinnings, coupled with a lack of ready-to-use computational tools, can make it difficult for researchers to engage with, test, and build upon these new models.

This paper addresses this challenge directly by introducing a comprehensive, self-contained, and interactive Jupyter notebook implementation of the UBP framework, version 3.6.2 (Computational Grammar Integration). The primary contribution of this work is to provide a stable, zero-dependency, and easily distributable tool that allows any researcher with a standard Python environment to conduct experiments within the UBP framework.

To validate the utility and scientific validity of this notebook, I present a detailed case study: a *Hydro-Informational Twinning Analysis*. This study investigates the informational costs associated with the phase transitions of water (H_2O) and the formation of the hydronium ion (H_3O^+). The results of this study not only demonstrate the notebook's capabilities but also provide compelling, empirically derived insights that support the core principles of the UBP. This paper documents the notebook's architecture, the case study's methodology and findings, and serves as an invitation to the scientific community to utilize this tool for further research.

2 Background: The Universal Binary Principle (UBP)

The Universal Binary Principle is a computational framework grounded in an *Information-First* worldview. It posits that information, not matter or energy, is the fundamental substrate of reality. In this model, the universe is a deterministic system of interacting binary states (toggles) within a high-dimensional "bitfield" - the virtual space of the UBP system, and all physical phenomena emerge from the geometric and computational rules governing these interactions.

Version 3.6 of the UBP introduces several key concepts that are critical to understanding the presented work:

2.1 Coherence-Native Computation

Unlike traditional computing where numerical precision is the primary measure of quality, the UBP operates on the principle of **coherence**. Every value in the system is a `CoherenceState`, an object that encapsulates not only a numerical magnitude but also its informational quality. This quality is quantified by the **Non-Random Coherence Index (NRCI)**, a measure of a state's fidelity to the underlying geometric structure of the substrate. In UBP, NRCI is not a post-hoc metric but is actively tracked and updated *during* computation.

2.2 Computational Grammar

The UBP 3.6 framework introduces a **Computational Grammar**, which defines a set of 10 primitive operators that form a closed algebra. These operators are not arbitrary conventions but are posited to be geometrically necessary stable states within the informational substrate. All other complex operations are compositions of these primitives. This grammatical structure allows for a complete audit trail of any computation, tracked via the `operator_sequence` attribute of a `CoherenceState`.

2.3 Geometric Constants and Y-Scaling

The UBP is defined by a set of fundamental geometric constants derived from first principles. The most critical of these is the **Y-constant** ($\mathrm{Y} = \pi/(\pi^2 + 2) \approx 0.2646$), which represents a fundamental geometric resonance in the substrate. The process of *Y-refinement* is a core operation that reflects the system's intrinsic self-correcting nature. Its inverse, *Y-inverse* ($1/\mathrm{Y} \approx 3.7782$), represents the informational cost of an observation.

2.4 The Nine Realms of Reality

The UBP framework is designed to be a complete and unified model, applicable across all scales of reality. It defines nine distinct **Physical Realms**, from the quantum to the cosmological, each governed by the same fundamental principles but operating under different parametric regimes. The interactive notebook presented in this paper includes complete implementations for all nine realms, enabling cross-domain research.

2.5 Framework Note

The Coherence-Native Computation, Computational Grammar, Geometric Constants, Y-Scaling, NRCI and The Nine Realms of Reality and more have all been extensively tested and developed in previous studies, this document doesn't aim to provide full explanations for all aspects of the UBP theory and system - you can find the documentation for deeper explanations in the UBP_Repo [1]

3 The UBP 3.6 Interactive Notebook

To lower the barrier to entry for UBP research, I developed a self-contained Jupyter notebook that encapsulates the entire UBP 3.6.2 framework. The design philosophy was to create a *zero-dependency*, *patch-free*, and *fully-validated* research tool.

3.1 Architecture and Design

The notebook is structured into cells - executable code cells and markdown cells for documentation. The core of the notebook consists of 27 cells, each containing a complete UBP 3.6 module. This modular-within-a-monolith design ensures that the entire system is loaded into the runtime environment sequentially, preventing namespace conflicts and ensuring that all dependencies are resolved internally.

Key architectural features include:

- **Zero External Dependencies:** The entire system is implemented in pure Python, using only standard libraries such as `math`, `typing`, and `dataclasses`. This eliminates the need for any `pip` installations and guarantees that the notebook will run in any standard Python 3.7+ environment, including restrictive ones.

- **Sequential Loading:** The modules are ordered to ensure that foundational classes like `CoherenceState` are defined before being referenced by other modules. This resolves dependency issues that arose in earlier, patch-reliant versions.
- **Integrated Documentation:** Each code cell is preceded by a markdown cell that explains the purpose of the module, its key functions, and its role within the UBP framework. This makes the notebook a learning tool as well as a research instrument.
- **Validation and Case Study:** The notebook concludes with a system validation cell and the comprehensive water study detailed in Section 4, providing users with an example of how to use the system for real research.

3.2 Accessibility and Distribution

The final notebook is available on both Google Colab [2] and GitHub [3], ensuring maximum accessibility. The public Colab link allows researchers to begin experiments, without any setup. The GitHub repository provides a version-controlled artifact for integration into other projects.

4 Computational Collaboration: Leveraging Gemini for UBP Research

The public UBP validation notebook, hosted on Google Colaboratory, is designed to facilitate rapid, interactive research through a collaborative model with Gemini, a large language model. This approach democratizes access to complex computational physics models, allowing researchers to focus on hypothesis generation and analysis rather than debugging code.

4.1 Protocol for Developing New Studies

To conduct new experiments or develop follow-up studies within the notebook environment, the user should adhere to the following three-step protocol:

1. **State the Hypothesis or Intent:** Clearly articulate the desired modification to the UBP system.
   - **Example Intent:** "I want to test the influence of the Observer Framework's cost by reducing the `Y_INVERSE` constant by 10%."
   - **Example Hypothesis:** "If the bias field is reversed to `(-0.03, -0.03)`, the storm center will move to the top-left quadrant."
2. **Request AI Implementation:** Instruct the Gemini model directly to write and execute the necessary Python code based on the stated intent, creating a new cell within the notebook's environment at the bottom of all the cells. The model is capable of understanding and manipulating the UBP framework's core classes (`CoherenceState`, `WeatherRealm`).
3. **Analyze the Outcome:** Review the generated numerical outputs (e.g., the Storm Tracking table) and the final conclusions to determine if the hypothesis was supported. This restores the user's role to the core scientific task of **analysis and inference**, significantly accelerating the research cycle.

This collaborative structure ensures that even computationally complex studies on concepts like **Anisotropic Flow** or **Observer Cost** can be tested and validated quickly by any researcher without requiring deep expertise in the underlying Python implementation.

5 Case Study: Hydro-Informational Twinning Analysis

To test functionality and to illustrate use of the UBP 3.6 notebook as a viable research tool, I conducted a *Hydro-Informational Twinning Analysis*. This study investigates the informational properties of water (H_2O) across its phase transitions and compares them to the formation of the hydronium ion (H_3O^+). The objective was to quantify and compare the informational cost of phase disorder versus topological reconfiguration from an Information-First perspective.

5.1 Methodology

The study was conducted entirely within the UBP 3.6 notebook and proceeded in seven distinct phases to show how iteration can develop during a study and feedback into it for further insights and fidelity.

5.1.1 State Initialization

Four molecular states were defined as initial `CoherenceState` objects. Each state was assigned an informational **D-value**, a dimensionless parameter representing the state's intrinsic complexity and disorder. These D-values are heuristic inputs that seed the simulation, representing the informational "cost" to manifest that state. The values were chosen to reflect the increasing disorder and complexity from solid ice to the pyramidal topology of hydronium.

- **H_2O (Ice):** D-value = 0.05 (baseline ordered state)
- **H_2O (Liquid):** D-value = 0.20 (increased disorder)
- **H_2O (Vapor):** D-value = 0.85 (high disorder)
- **H_3O^+ (Ion):** D-value = 1.20 (high disorder + topological complexity)

Each state's initial NRCI was calculated by applying the informational cost (D-value) to a perfect coherence state using the `degrade_by()` method.

5.1.2 Analysis Phases

The simulation followed a seven-phase protocol using study data as system testing:

1. **Phase Transitions:** Initial NRCI values for the four states were calculated.
2. **Molecular Complexity:** The hydronium state was established.
3. **Geometric Error Correction:** The `restore_coherence()` function, using a dodecahedral pattern, was applied to each state to quantify potential coherence recovery.
4. **Informational Entropy Analysis:** A UBP-specific measure of informational entropy (S_{info}) was calculated for each state based on its NRCI.
5. **Cross-Phase Coherence Matrix:** The pairwise informational distance (ΔNRCI) between all four states was computed.
6. **Resonance Pattern Detection:** The ratios of D-values were analyzed to detect nonlinearities in informational cost.
7. **HexDictionary Persistence:** The final states were stored in the `HexDictionary`, a content-addressable storage system, to validate data persistence.

5.2 Results

The study yielded several key quantitative results, summarized below.

5.2.1 State Coherence and Informational Cost

The initial NRCI values directly reflect the informational cost (D-value) of each state. As shown in Table 1, increasing disorder and complexity lead to a monotonic decrease in coherence.

Table 1: Primary Results of the Hydro-Informational Twinning Analysis

State	D-Value	Initial NRCI	S_{info} (Info. Entropy)
H_2O (Ice)	0.05	0.999996999250	7.50×10^{-10}
H_2O (Liquid)	0.20	0.999996996248	3.75×10^{-9}
H_2O (Vapor)	0.85	0.999996983455	1.65×10^{-8}
H_3O^+ (Ion)	1.20	0.999996978171	2.18×10^{-8}

5.2.2 Geometric Correction Efficacy

The application of dodecahedral geometric error correction resulted in a small but consistent improvement in coherence for all states, demonstrating the system's self-correcting nature. The average NRCI improvement was $\approx +4.80 \times 10^{-15}$, indicating that even stable states possess a residual geometric "error" that can be restored.

5.2.3 Informational Cost of State Transitions

The cross-phase coherence matrix revealed the informational cost of transitioning between states. The key comparison is between the cost of phase disorder (liquid to vapor) and the cost of topological reconfiguration (liquid to hydronium).

- Cost of Vaporization (Disorder): ΔNRCI (Liquid $\rightarrow$ Vapor) $= 1.28 \times 10^{-8}$
- Cost of Ionization (Topology): ΔNRCI (Liquid $\rightarrow H_3O^+$) $= 1.81 \times 10^{-8}$

The analysis shows that the formation of the topologically complex hydronium ion is approximately **1.41 times more informationally expensive** than the phase transition to a highly disordered vapor state. This is the interesting finding of the study.

6 Discussion

The results of the Hydro-Informational Twinning Analysis provide strong empirical support for the core tenets of the UBP framework. The central finding—that topological reconfiguration is informationally more expensive than phase disorder—is a non-trivial prediction of the Information-First model. In a traditional thermodynamic view, one might expect the high-entropy vapor state to represent the greatest departure from the baseline. However, the UBP framework, by treating geometry and information as primary, correctly predicts that the creation of a new topological structure (the pyramidal H_3O^+) requires a greater expenditure of informational resources.

This result has significant implications. It suggests that the complexity of a system, from a UBP perspective, is not merely a function of its randomness or disorder, but of its underlying geometric and topological structure. The informational cost is a measure of what is required to *specify* a state within the substrate, and complex geometries require more information to specify than simple, disordered ones.

Furthermore, the successful application of geometric error correction, even on stable states, reinforces the UBP concept of a dynamic, self-correcting substrate. The minute but measurable increase in NRCI suggests that all physical states are in a constant process of coherence maintenance, and that geometric integrity is a fundamental property of reality.

The successful execution of this entire study within a single, self-contained notebook validates the primary goal of this project: to make UBP research accessible. Researchers can now replicate this study, modify its parameters, and design new experiments without needing to set up a complex software environment or master the framework's entire codebase from scratch.

7 Conclusion

I have provided a stable, accessible, and fully-validated interactive notebook for conducting scientific research using the Universal Binary Principle (UBP) v3.6. By consolidating the entire framework into a zero-dependency Jupyter notebook, I have hopefully lowered the barrier to entry for researchers wishing to explore this Information-First model of reality. The notebook's architecture, technical enhancements, and integrated documentation make it a powerful tool for both learning and active research.

The successful completion of the Hydro-Informational Twinning Analysis serves as a robust validation of the notebook's capabilities. The study's key finding — that **topological complexity is informationally more expensive than phase disorder** — provides compelling evidence for the UBP's core principles and demonstrates the unique insights that can be gained from an Information-First perspective.

This work represents a critical step in transitioning the UBP from a theoretical framework to a practical, testable scientific model. I invite the scientific community to use this tool to validate the findings, conduct new research, and contribute to the ongoing exploration of our informational universe. The public availability of the notebook on Google Colab and GitHub is intended to foster a collaborative and open research environment.

References

[1] E. R. A. Craig, *UBP_Repo: The Universal Binary Principle Repository*. GitHub. https://github.com/DigitalEuan/UBP_Repo

[2] E. R. A. Craig, *UBP Validation Study: Anisotropic Flow Test (Part 6)*. Google Colaboratory. https://colab.research.google.com/drive/1618aclLyllZJAugInXa3pq-dnVzCz2ho?usp=sharing

[3] E. R. A. Craig, *UBP 3.6 Notebook File (Anisotropic Flow)*. GitHub Repository. https://github.com/DigitalEuan/UBP_Repo/tree/main/UBP_Weather_1.ipynb